Brian Keller

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SUMMARY

Prospective ML/AI engineer with a biomedical engineering foundation and hands-on experience building and deploying deep learning models on multi-modal datasets. Comfortable across the full pipeline from data collection and preprocessing to model training, evaluation, and deployment.

TECHNICAL SKILLS

Machine Learning & AI: Deep Learning, Computer Vision, PyTorch, scikit-learn, ResNet, YOLOv8, CNNs, model training & evaluation, AdamW / Adam optimization, label smoothing, data augmentation, static optimization, model deployment (FastAPI) musculoskeletal simulation (OpenSim)

Data & Analytics: Pandas, NumPy, data preprocessing, feature engineering, data visualization, relational databases, large-scale multi-modal dataset management

Programming: Python, CircuitPython, Java, MATLAB, C#, Git

Tools & Engineering: Vicon motion capture, Materialise Mimics, Fusion 360 (CAD), additive manufacturing, manual milling, manual lathing, CNC machining, FeatureCAM

EDUCATION

University of Utah, John and Maria Price College of Engineering Salt Lake City, UT
BS/MS Biomedical Engineering, Minor in Computer Science Dec 2026
Graduate Certificate, Deep Learning & AI Robotics Dec 2026
Presidential Scholar · Daniels Scholar · Tau Beta Pi Member · Dean's List 2021–26 · GPA 3.96

WORK EXPERIENCE

Graduate Research Assistant — *Foot & Ankle Mechanics and Morphology (FAMM) Lab* May 2024 – Present

- Designed scalable preprocessing workflows for multi-modal biomechanics data (Vicon motion capture, ground reaction forces), eliminating manual formatting and standardizing inputs for deep learning pipelines.
- Processed patient weight-bearing CT scans with 3D modeling software to quantify bone morphology and joint kinematics, and managed large motion-capture and force datasets optimized for model training.
- Operated, maintained, and trained others on a multi-camera Vicon optical motion capture system to ensure precise, repeatable data collection.

Lift & Ride Operator — *Snowbird Mountain Resort* May 2021 – Apr 2025

- Directed a crew of operators to monitor guest safety, enforce procedures, and maintain smooth, uninterrupted lift operations.
- Trained and mentored new hires while inspecting and documenting equipment issues to ensure safe performance.

PROJECTS

OpenSim Deep Learning Project — *University of Utah, FAMM Lab* Jun 2025 – Present

- Built end-to-end pipelines for and trained deep learning models (PyTorch) that predict lower-limb muscle and joint contact forces from ground reaction force data alone across older- and younger-adult populations, enabling live feedback for a robotic controller.
- Generated ground-truth force labels by simulating each experimental trial in OpenSim with a custom static optimization routine, producing muscle force estimates across all subjects for model training and validation.

Automated Nursery Inventory Viability System — *University of Utah / BloomLogic* Jan 2026 – Present

- Co-developed an end-to-end computer vision pipeline (YOLOv8 tray segmentation → ResNet-18 grid classification → ResNet-18 per-cell classification) that automates seedling-tray viability from mobile photos, achieving 96.74% per-cell accuracy and a tray-level viability MAE of 6.01 percentage points (Pearson r = 0.942).
- Trained and evaluated ResNet-18 models in PyTorch / timm on a 9,155-image labeled dataset using AdamW, label smoothing, and targeted augmentation; deployed the system as a FastAPI service with a mobile iOS capture app at 1.65 s median inference over a 1,037-image production batch.
- Conducted on-site interviews with commercial nursery staff to translate qualitative grading criteria into implementable technical specifications.

Esophageal SpO_2 Monitor Capstone — *University of Utah* Jan – Nov 2024

- Led software development on a multidisciplinary team building a novel esophageal SpO_2 monitor, developing embedded firmware in CircuitPython and a user-facing UI for the integrated device.
- Supported sensor integration and component selection and contributed to the verification/validation documentation required to meet FDA and ISO 13485 standards.

LEADERSHIP

Eagle Scout (2018) · Students for the Wasatch, Merchandise Coordinator (2022–23) · Car Club at the U, Officer (2025-26)